

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Selected System: A Personalized News Feed Recommender, such as the Twitter/X or TikTok algorithm.

Task 1: PEAS Specification

PEAS Element	Specification
Performance Measures	User engagement rate; watch time or reading time; content relevance score; user satisfaction signals; content diversity; reduction of harmful, spammy, or misleading content.
Environment	A social media platform containing user profiles, posts, videos, hashtags, captions, comments, ads, trending topics, and platform rules. External factors include changing trends, current events, creator activity, user mood, and language/location context.
Actuators	Feed ranking system; recommended posts/videos display; content filtering and hiding tools; notification system; account/topic/hashtag suggestions; advertisement placement system.
Sensors	Likes, comments, shares, reposts; watch time and scroll speed; searches and clicks; followed, muted, and blocked accounts; user language and location; hashtags, captions, keywords, video/audio features; reports and "not interested" feedback.

Task 2: Environment Classification

Dimension	Choice	Justification
Observability	Partially Observable	The system can observe clicks and watch time, but it cannot directly know the user's true mood, intent, or private preferences.
Agents	Multi-agent	Users, creators, advertisers, moderators, and other recommendation processes interact on the same platform.
Determinism	Stochastic	A user may respond differently to similar content depending on time, mood, context, and competing posts.
Episode Type	Sequential	Each recommendation influences later user behavior, collected signals, and future recommendations.
Change Over Time	Dynamic	New posts, trends, comments, and user actions appear continuously while the agent is operating.
State/Action Space	Continuous	Preference scores, ranking values, watch time, and engagement probabilities change gradually over time.

Task 3: Critical Analysis

1. Greatest technical challenge: The most difficult dimension is partial observability. The recommender receives behavioral clues such as clicks, watch time, likes, and shares, but these are only indirect signals. A long watch time may mean the user enjoyed the content, or it may mean the user was confused, shocked, or simply paused the app. Because the real reason is hidden, engineers must design models that infer preferences carefully instead of blindly maximizing engagement.

2. Utility function: One possible utility function is $U = 0.4 \cdot \text{Engagement} + 0.3 \cdot \text{Relevance} + 0.2 \cdot \text{Diversity} - 0.1 \cdot \text{Harmful_Content_Risk}$. This balances keeping users interested with showing relevant, varied, and safer content. If the harmful-content weight is doubled to 0.2, the agent becomes more cautious: it will avoid risky or misleading posts more often, even if some of those posts would have produced high engagement.

Structured Representation

The following JSON object stores the PEAS specification, environment classification, and utility function in a machine-readable form.

```
{
  "system_name": "Personalized News Feed Recommender",
  "peas": {
    "performance_measures": [
      "user_engagement_rate",
      "watch_time_or_reading_time",
      "content_relevance_score",
      "user_satisfaction",
      "content_diversity",
      "reduction_of_harmful_or_misleading_content"
    ],
    "environment": "Social media platform with profiles, posts, videos, ads, trends, and rules.",
    "actuators": [
      "feed_ranking_system",
      "recommended_posts_display",
      "content_filtering_tool",
      "notification_system",
      "account_and_hashtag_suggestions",
      "advertisement_insertion_system"
    ],
    "sensors": [
      "likes_comments_shares_and_reposts",
      "watch_time_and_scroll_speed",
      "search_history_and_clicked_posts",
      "followed_muted_and_blocked_accounts",
      "user_language_and_location",
      "hashtags_captions_keywords_and_media_features",
      "reports_not_interested_clicks_and_hidden_posts"
    ]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Real user mood, intent, and private preferences are hidden."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "Users, creators, advertisers, moderators, and systems interact."
    },
    "determinism": {
      "choice": "Stochastic",
      "justification": "User reactions are uncertain and can change for similar content."
    },
    "episodic_sequential": {
      "choice": "Sequential",
      "justification": "Each recommendation affects later behavior and recommendations."
    },
    "static_dynamic": {
      "choice": "Dynamic",
      "justification": "New content, trends, and user actions are constantly changing."
    },
    "discrete_continuous": {
      "choice": "Continuous",
      "justification": "Interest, engagement, relevance, and ranking scores change over time."
    }
  },
  "utility_function": "U = 0.4*Engagement + 0.3*Relevance + 0.2*Diversity - 0.1*Harmful_Content_Risk"
}
```